$$y_p = \frac{1}{f(D^2)} \sin ax \qquad D^2 \to -a^2$$

$$= \frac{1}{f(-a^2)} \sin ax$$

if $f(-a^2) = 0$,

then

$$= \frac{1}{f'(D^2)} \sin ax$$

$$= x \frac{1}{f'(-a^2)} \sin ax$$

$$\frac{d^2y}{dx^2} + y = \sin 2x \quad -(1)$$

$$\boxed{\frac{d^2y}{dx^2} + y = \cos 2x}$$

Soln: let $Dy = \frac{dy}{dx}$, $D = \frac{d}{dx}$

$D^2y = \frac{d^2y}{dx^2}$, $D^2 = \frac{d^2}{dx^2}$

$\therefore (1) \Rightarrow D^2y + y = \sin 2x$

$\Rightarrow (D^2+1)y = \sin 2x \quad -(2)$

let $y = e^{mx}$ be the possible solution of $(D^2+1)y = 0 \quad -(3)$

$\therefore$ (3) from eqn (3) we can write $(D^2+1)e^{mx} = 0$

$\Rightarrow m^2 e^{mx} + e^{mx} = 0$

$\Rightarrow (m^2+1) e^{mx} = 0$

$\Rightarrow m^2+1=0 \quad [\because e^{mx} \neq 0]$

$\Rightarrow m^2 = -1$

$\Rightarrow m^2 = i^2$

$\Rightarrow m = \pm i$

$\therefore y_c(x) = A\cos x + B\sin x$

Now, $y_p = \dfrac{1}{D^2+1} \sin 2x$

$= \dfrac{1}{-2^2+1} \sin 2x$

$= \dfrac{1}{-3} \sin 2x$

$= -\frac{1}{3} \sin 2x$

$\therefore$ General solution of the given differential eqn. is $y(x) = y_c(x) + y_p(x)$

$= A\cos x + B\sin x - \frac{1}{3}\sin 2x.$

3

$y_p = \frac{x e^{ax}}{f(D)}$ $\quad$ $y_p = \frac{1}{f(D)} e^{ax} \sin bx$

$= e^{ax} \frac{1}{f(D+a)} x$ $\quad$ $= e^{ax} \frac{1}{f(D+a)} \sin bx$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2 e^{2x}$

Solⁿ: $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2 e^{2x}$ — (1)

let ~~$y = e^{mx}$~~ $y = e^{mx}$ be the possible solution of

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ — (2)

$d \therefore (2) \Rightarrow \frac{d^2}{dx^2}(e^{mx}) - 4\frac{d}{dx}(e^{mx}) + 4e^{mx} = 0$

$\Rightarrow m^2 e^{mx} - 4me^{mx} + 4e^{mx} = 0$

$\Rightarrow (m^2 - 4m + 4)e^{mx} = 0$

$\Rightarrow m^2 - 4m + 4 = 0 \quad [\because e^{mx} \neq 0]$

$\Rightarrow m = 2, 2$

$y_c(x) = \cancel{c + e^m} (c_1 + c_2 x) e^{2x}$

let $D = \frac{d}{dx}$ and $D^2 = \frac{d^2}{dx^2}$

$\therefore$ (1) $\Rightarrow D^2y - 4Dy + 4y = x^2e^{2x}$

$\Rightarrow (D^2 - 4D + 4)y = x^2e^{2x}$

$\therefore y_p(x) = \frac{1}{D^2-4D+4} x^2 e^{2x}$

$= \frac{1}{(D-2)^2} x^2 e^{2x}$

$= \cancel{e^{2x} \frac{1}{(D-2)^2}}$

$= e^{2x} \frac{1}{(D+2-2)^2} x^2$

$= e^{2x} \frac{1}{D^2} x^2$

$= \frac{1}{12} x^4 e^{2x}$

$\therefore$ G S, $y(x) = y_c + y_p$

$= (c_1 + c_2 x)e^{2x} + \frac{1}{12} x^4 e^{2x}$

3

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \sin x$$

Soln: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \sin x \quad -(1)$

Let $y = e^{mx}$ be the possible solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0 \; -(2)$

Then (2) $\Rightarrow \frac{d^2}{dx^2}(e^{mx}) - 2\frac{d}{dx}(e^{mx}) + e^{mx} = 0$

$\Rightarrow m^2 e^{mx} - 2m e^{mx} + e^{mx} = 0$

$\Rightarrow (m^2 - 2m + 1) e^{mx} = 0$

$\Rightarrow m^2 - 2m + 1 = 0 \quad [\because e^{mx} \neq 0]$

$\Rightarrow m = 1, 1$

$\boxed{\begin{array}{l} m^2 - 2m + 1 = 0 \\ \Rightarrow (m-1)^2 = 0 \end{array}}$

$\therefore y_c(x) = (c_1 + c_2 x) e^x$

Let $\frac{d}{dx} = D$ and $D^2 = \frac{d^2}{dx^2}$

$\therefore (1) \Rightarrow D^2 y - 2Dy + y = e^x \sin x$

$\Rightarrow (D^2 - 2D + 1) y = e^x \sin x$

$\Rightarrow y_p = \frac{1}{D^2 - 2D + 1} e^x \sin x$

$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 1} \sin x$

$= e^x \frac{1}{D^2 + \cancel{2D} + \cancel{1} - \cancel{2D} - \cancel{2} + \cancel{1}} \sin x$

$= e^x \frac{1}{D^2} \sin x$

$= -e^x \sin x$

$\therefore y(x) = y_c + y_p = (c_1 + c_2 x) e^x - e^x \sin x$

6

Cauchy Euler equation

$$x^3\frac{d^3y}{dx^3}+3x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+2y=0 \quad -(1)$$

Let $x=e^z \Rightarrow z=\ln x$

$$\therefore \frac{dz}{dx}=\frac{1}{x}$$

Now $\frac{dy}{dx}=\frac{dy}{dz}\frac{dz}{dx}=\frac{1}{x}\frac{dy}{dz} \Rightarrow x\frac{dy}{dx}=\frac{dy}{dz}=Dy \quad \left[D=\frac{d}{dz}\right]$

$$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right)=\frac{1}{x}\frac{d^2y}{dz^2}\frac{dz}{dx}-\frac{1}{x^2}\frac{dy}{dz}$$

$$=\frac{1}{x^2}\frac{d^2y}{dz^2}-\frac{1}{x^2}\frac{dy}{dz}$$

$$\Rightarrow x^2\frac{d^2y}{dx^2}=\frac{d^2y}{dz^2}-\frac{dy}{dz}=D^2y-Dy=D(D-1)y$$

Similarly $x^3\frac{d^3y}{dx^3}=D(D-1)(D-2)y$

$x=e^z$

$x^{-2}=e^{-2z}$

$\Rightarrow \frac{1}{x^2}=e^{-2z}$

$\therefore$ (1) Can be written as,

$$D(D-1)(D-2)y+3D(D-1)y-2Dy+2y=0$$

$$\Rightarrow D(D^2-3D+2)y+3(D^2-D)y-2Dy+2y=0$$

$$\Rightarrow (D^3-3D+2)y=0 \quad -(2)$$

Let $y=e^{mz}$ be the possible solution of (2).

$$(D^3-3D+2)e^{mz}=0$$

$$\Rightarrow (m^3-3m+2)e^{mz}=0$$

$$\Rightarrow m^3-3m+2=0$$

$\Rightarrow m=1,1,-2$

$\therefore y(z)=(c_1+c_2z)e^z+c_3e^{-2z}$

$y(x)=(c_1+c_2\ln x)x+c_3\frac{1}{x^2}$